LAB 2 REPORT

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Lab Title: Functions in Pyhton

Theory: Functions in Python are reusable blocks of code that perform a specific task.They help organize programs, reduce repetition, and make code easier to understand.Functions are created using the `def` keyword and can take inputs and return outputs.

```
# TASK 1
def factorial(num):
    fact = 1
    for i in range(1, num + 1):
        fact = fact * i
    return fact
def permutation(n, r):
    return factorial(n) // factorial(n - r)
def combination(n, r):
    return factorial(n) // (factorial(r) * factorial(n - r))
print("=== Permutation and Combination Calculator ===\n")
n = int(input("Enter value of n: "))
r = int(input("Enter value of r: "))
if r > n or n < 0 or r < 0:
    print("Error: r cannot be greater than n or negative values not allowed!")
else:
    print(f"\nPermutation P({n}, {r}) = {permutation(n, r)}")
    print(f"Combination C({n}, {r}) = {combination(n, r)}")
```

Output:

```
Enter value of n: 2
Enter value of r: 1

Permutation P(2, 1) = 2
Combination C(2, 1) = 2

Process finished with exit code 0
```

```
# TASK 2
larger = lambda a, b: a if a > b else b
def print_table(num, start, end):
```

```python
    print(f"\nTable of {num} from {start} to {end}:")
    for i in range(start, end + 1):
        print(f"{num} x {i} = {num * i}")
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
large_num = larger(a, b)
print(f"Larger number is: {large_num}")
start = int(input("Enter start of table range: "))
end = int(input("Enter end of table range: "))
print_table(large_num, start, end)
```

Output:

```
Enter first number: 3
Enter second number: 4
Larger number is: 4
Enter start of table range: 1
Enter end of table range: 11

Table of 4 from 1 to 11:
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
4 x 11 = 44

Process finished with exit code 0
```

```python
# TASK 3
to_upper = lambda s: s.upper()
def invert(text):
    print("Reversed Uppercase String:", text[::-1])
text = input("Enter a string: ")
upper_text = to_upper(text)
print("Uppercase:", upper_text)
invert(upper_text)
```

Output:

```
Enter a string: sohaib
Uppercase: SOHAIB
Reversed Uppercase String: BIAHOS

Process finished with exit code 0
```